Analysis & Design Document

A Guide for its Preparation…

Introduction

- ADD is *a continuation of* the project specification document (*PSD*)
- ***Goal:*** presenting the analysis and design of the project.
- Focuses on *two different types* of projects:
  - ***Software oriented projects***
  - ***Academia oriented projects***
- Type of project to determine by advisor !!!

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
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Introduction

- Problem Description and Motivation (revised & extended from PSD)
  - What is the *problem*?
  - Why is it *worth solving* it?
- Scope of the Project (revised from PSD)
  - What is *in* and what is *beyond* the project?
  - What to *use* and what to *develop*?
- Definitions, acronyms, and abbreviations

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Literature Survey

- *Refine* your *literature survey* in PSD with additional literature check
  - *in the direction of your advisor's opinions and the comments of your PSD reviewers*
  - *by adding* any possibly *new* or *just missed work* related to your problem/work.
  - to justify your work by pointing to its new/different aspects.
- Any new *work critically similar* to your project *you* should *rigorously discuss* in the section «*related work*».

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Project Requirements

- *Functional* Requirements
  - *Description of system functionality*
    - System's reaction to particular inputs and
    - System's behaviour in particular situations.
- *Non-functional (NF)* Requirements
  - *Properties* of the system
  - *Constraints on the*
    - *services or functions* offered by the system such as timing constraints,
    - *development process, standards*, etc.
  - Must be ***verifiable*** (i.e., use a testable *measure*).

NF Requirements & measures

- *Performance* *(throughput, response time, screen refreshing time)*
- ***Reliability*** *(ROFOC, MTTF)(rate of failure occurrence, mean time to failure)*
- *Usability* *(Ease of use, # of help frames, warn./err. messages, size of user's guide) (# of errors made per unit time/training time)*
- ***Security*** *(only for networking systems; number of unauthorized accesses to the system)*
- *Maintainability* *(code modularity, rate of bug repair per unit time)*
- ***Portability/platform-independence*** *(% target dependent statements)*
- *Scalability* *(# of running components, max. # of users with system up)*

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System Design

- *UML Use case diagrams* for main use cases
  - to specify *scenarios / use cases* of the system
- *UML Class and/or Database ER diagrams…*
  - show building blocks of your system
  - represent *a static view* of your model
  - describe attributes and behavior for targeted operations
- *UML Sequence diagrams…*
  - describe a sequence of *interactions among actors and use cases* to attain the goal of the specified behaviour.
- *User Interface (a preliminary version)*
- Test Plan

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Software Architecture

- Display
  - the system's *structure* and *static aspects* such as *subsystems/modules* using *block diagrams*, and other *visual aids*.
  - its *dynamic perspective* such as *data* and *control flow* using *state machines* *(especially fit for real-time systems)*

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Tasks Accomplished

- *Current state* of the project (implementation and preliminary results)
- *Task Log* (information about meetings and activities, including date, short description and hours)
- *Task Plan with Milestones* (clear and well-defined descriptions of the *work that must be completed* before predetermined check points, illustrated by *Gantt chart*)

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Related Work

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  - to justify your work by pointing to its new/different aspects.
- *Rigorously discuss* any newly found *work/method* essential/*comparable to your work* in «*related work*».

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Preliminaries

- Provide, if any, necessary *fundamental or background concepts* you need know to accomplish your main tasks in this project.

- Be rigorous in including all necessary *fundamentals* etc. you will be exploiting.

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System Design (Methodology)

- *System Model (block diagrams for static & state machines for dynamic perspective, other visual aids welcome)*
- *Flow charts and/or pseudocode of proposed algorithms*
- *Performance assessment criteria, comparison metrics (provide detailed explanation, the more unambiguous the better!!!)*
- *Data sets or benchmarks (provide detailed explanation)*

System Design (Methodology)

- *Flow charts and/or pseudocode of proposed algorithms*
  - Give enough details!
  - Consider an *individual* pseudocode per module!
- *Performance assessment criteria, comparison metrics*
  - *select one of the following (or some other) appropriate for your work to compare your results with the state of the art (SOTA) (i.e., newest work in literature)*
    - *Adjusted Rand Index (ARI),*
    - *F(1)-score,*
    - *precision/recall or*
    - *other standard accuracy criteria* such as *sensitivity*, *specificity*, *false positive/negative rates*, confusion matrix etc.)

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Experimental Study

- *Experiment setup*
  - For each experiment that you conduct, specify
    - what *idea(s)* this experiment *supports*
    - *how it works and it supports the relevant idea*
    - any *parameter values, necessary system specs*
- *Results*
  - *Show* your results *using curves, tables, figures and other visual aids.*
- *Detailed discussion*
  - Analyze and clarify *all aspects of the results*.
  - *Compare your results* with SOTA and show!!!

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Generative AI tool “Incidence”

- *All PSDs I read point undeniably to help received from some text generating AI tool.*
- *That is acceptable as long as you take a full responsibility of what is written.*
  - *be fully knowledgeable of the text (each word and word combination) and*
  - *have in-depth understanding of the idea the document reflects*
  - *since **it literally is "your" document**.*
- *The author(s) of any document must be fully responsible for the reliability of their document meaning that they must be in full command of what is written*
  - *from the general idea*
  - *to each specific word or phrase chosen to exist in the document.*

Generative AI tool "Incidence"

- *If you have such command of your PSD that is perfect. If not, start **now** worrying!*
- *A document just like any work you do or any product you create reflects your style and directly becomes a hallmark of your character.*
- *Hence, make sure that you*
  - ***Only utilize the AI tool!***
  - ***Do never seem capable of what AI tool does for you which you cannot!***
  - ***Do never confine your capabilities to the tool's limits!***
  - ***Do not allow its mistakes to be your own mistakes!***
- *since you are responsible in all aspects (ethical, legislative and factual) for your output and not the tool that helps you!*

Thank you for your attention!